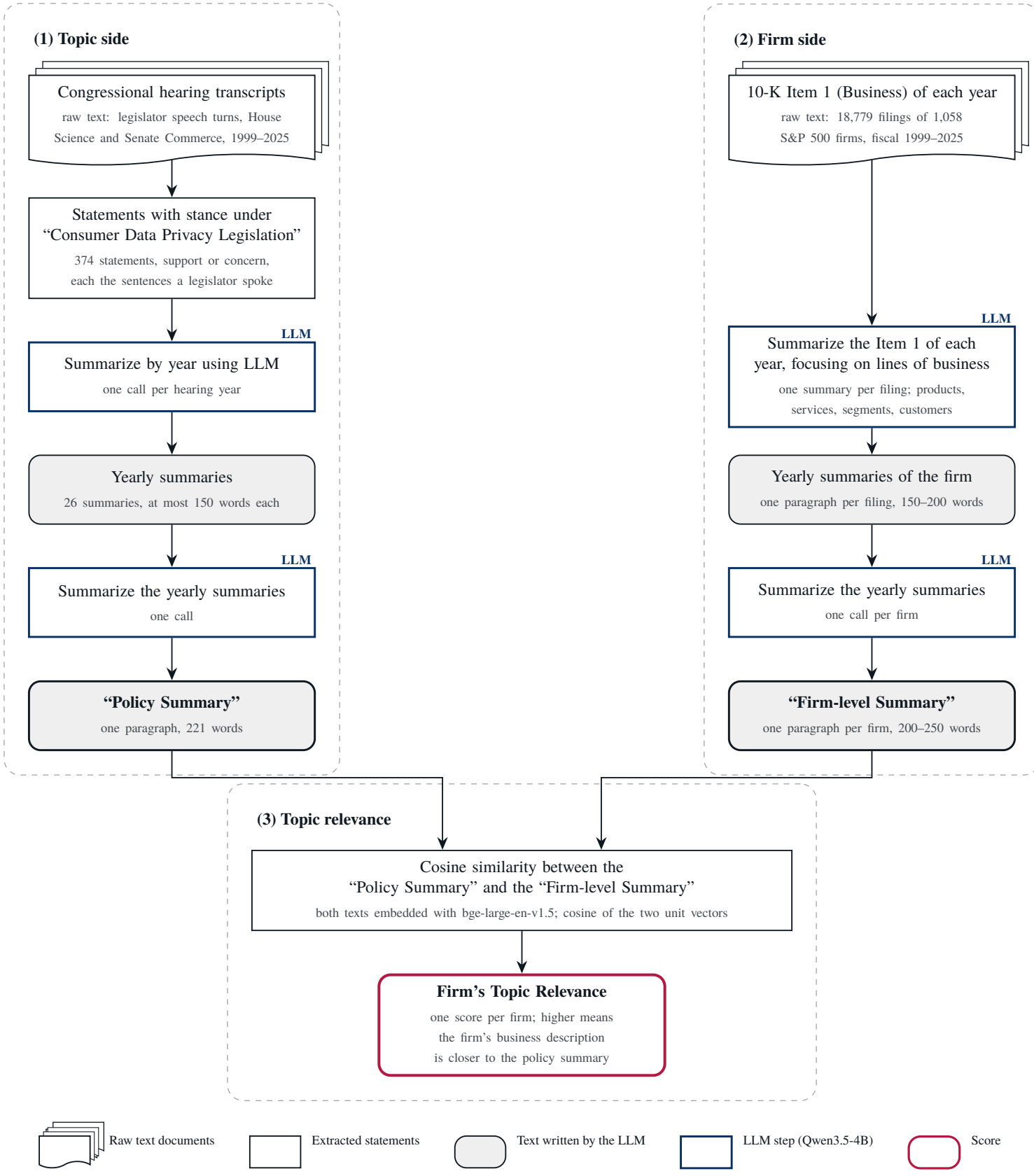

(1) Topic side
Congressional hearing transcripts
raw text: legislator speech turns, House Science and Senate Commerce, 1999–2025
Statements with stance under "Consumer Data Privacy Legislation"
374 statements, support or concern, each the sentences a legislator spoke
LLM
Summarize by year using LLM
one call per hearing year
Yearly summaries
26 summaries, at most 150 words each
LLM
Summarize the yearly summaries
one call
"Policy Summary"
one paragraph, 221 words
(2) Firm side
10-K Item 1 (Business) of each year
raw text: 18,779 filings of 1,058 S&P 500 firms, fiscal 1999–2025
LLM
Summarize the Item 1 of each year, focusing on lines of business
one summary per filing; products, services, segments, customers
Yearly summaries of the firm
one paragraph per filing, 150–200 words
LLM
Summarize the yearly summaries
one call per firm
"Firm-level Summary"
one paragraph per firm, 200–250 words
(3) Topic relevance
Cosine similarity between the "Policy Summary" and the "Firm-level Summary"
both texts embedded with bge-large-en-v1.5; cosine of the two unit vectors
Firm's Topic Relevance
one score per firm; higher means the firm's business description is closer to the policy summary
Raw text documents
Extracted statements
Text written by the LLM
LLM step (Qwen3.5-4B)
Score